

RGIPT

Subject: ai

Class: 12

Time Allowed: 45 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class: 12 Section: _____

Section A: Multiple Choice Questions

Choose the correct option. Each carries 1 mark.

1. [Easy] Which of the following best describes a limitation of character-level models? A) They capture semantic meaning directly. B) They require many steps to process long words, making them inefficient. C) They have the smallest possible vocabulary size. D) They avoid the need for embeddings. [1 Marks]
2. [Easy] Which algorithm is commonly used to construct subword vocabularies? A) WordPiece B) Byte Pair Encoding (BPE) C) TF-IDF D) LZW [1 Marks]
3. [Easy] Positional encoding in the original transformer relies on which mathematical functions? A) Linear functions B) Polynomial functions C) Sine and cosine waves D) Random vectors [1 Marks]
4. [Easy] What does a token embedding vector primarily represent? A) The token's position in the sentence B) A dense semantic representation of the token C) The frequency of the token in the corpus D) The character code of the token [1 Marks]

Section B: Short Answer Questions

Answer briefly. Each carries 2 marks.

1. [Moderate] Explain why a fixed word-level vocabulary creates an "unknown" problem and how subword tokenization mitigates this issue. [2 Marks]
2. [Challenging] Describe how sinusoidal positional encodings enable the model to generalize to sequence lengths longer than those seen during training. [2 Marks]
3. [Moderate] Summarize the combined role of embeddings and positional encodings in the transformer pipeline. [2 Marks]

End of Question Paper

Answer Key:

1. B
2. B
3. C
4. B
5. A fixed word-level vocabulary cannot represent words not seen during training, so unseen words become "unknown" tokens and the model loses meaning. Subword tokenization breaks rare or new words into known subword pieces, allowing the model to compose representations for unseen words from familiar parts, thus reducing the unknown problem.
6. Sinusoidal encodings assign each position a vector generated by sine and cosine functions of varying frequencies. Because these functions are continuous and periodic, the pattern for any position can be computed for arbitrarily large indices. The linear relationship between positions ($p+k$) and the encoding allows the model to infer relative distances even for positions beyond the training maximum, supporting extrapolation to longer sequences.
7. Embeddings convert discrete token IDs into dense high-dimensional vectors that capture semantic similarity. Positional encodings add a deterministic vector to each embedding that encodes the token's position in the sequence. Together, they provide the model with both "what" a token means and "where" it occurs, enabling the transformer to process language with meaning and order awareness.